Answer: A

31 A**:** $S_{1,}$ S_2 and S_3 closed: $R_{XY} = \dfrac{R}{2} + \dfrac{R}{3} = \dfrac{5R}{6}$

B: S_1 and S_3 closed but S_2 open : $R_{XY} = \dfrac{R}{2} + \dfrac{R}{2} = \dfrac{6R}{6}$

C: S_1 open but S_3 and S_2 closed: $R_{XY} = \dfrac{R}{2} + \dfrac{R}{2} = \dfrac{6R}{6}$

D: S_1 and S_3 open but S_2 closed: $R_{XY} = \dfrac{R}{2} + R = \dfrac{9R}{6}$

Answer: D

32 From the graph, of the 4 options only when the current = 0.1 A, is the sum of pd across